Richard Kim

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Education

University of California, San Diego Expected Dec. 2026
Bachelor of Science, Computer Engineering

- Relevant Coursework: Signals and Systems, Microelectronics, Digital Circuit Design, Computer Architecture, Algorithms, Data Structures, Systems Programming, Control Systems, Power Electronics, Semiconductor Physics

Experience

Software Engineer Intern
Y STEM and Chess – Boise, Idaho July 2026 – Present

- Built and operated an open-source SOC 2 GRC program using Probo (policies, controls, vendors, tasks, evidence) for an education tech platform
- Mapped SOC 2 Trust Services Criteria (Security + Confidentiality) to product systems (Azure, MERN, MongoDB, GitHub, Google Workspace, Discord, Quickbooks)
- Automated technical evidence collection with AuditKit (Axure SOC 2 scans) and custom scripts (GitHub PR exports, npm audit) for change management and vulnerability tracking

Student Intern
San Diego Supercomputer Center – La Jolla, CA Aug. 2026

- Configured and troubleshot Python and Linux-based development environments for participants at the CIML and SDSC Summer Institutes.
- Resolved real-time technical issues involving GitHub repositories, version control, and terminal usage during live software tutorials.
- Streamlined technical operations by diagnosing software issues and executing prompt problem-solving strategies.

Mechanical Team Member | Miramar Engineering Club
San Diego Miramar College – San Diego, CA Oct. 2022 – May 2024

- Designed and modeled custom mounting brackets for hovercraft lift fans, ensuring secure integration with the main chassis under operational loads
- Developed CAD designs for custom rocket components, refining geometries based on physical constraints, material properties, and aerodynamic considerations
- Iteratively prototyped and optimized bracket designs, improving structural integrity, manufacturability, and ease of assembly
- Collaborated with engineering team to evaluate design performance and implement improvements across multiple design revisions

Projects

Postura (Smart Posture Monitor) | Engineering Capstone github

- Developed C++ firmware for ESP32 microcontroller to acquire and filter spatial data from Time-of-Flight (ToF) sensor, implementing MQTT over Wi-Fi to publish real-time posture telemetry to a frontend web dashboard.
- Architected embedded hardware system, integrating the microcontroller, ToF sensor arrays, and power management components to ensure reliable data transmission and system stability.
- Designed and iterated mechanical enclosures using Fusion 360, optimizing the physical layout for seamless and comfortable integration of the electronics suite into a standard seat cushion.
- Conducted targeted user interviews with students and library staff to understand ergonomic habits and pain points, translating qualitative feedback into actionable hardware and software requirements.

AI Thermal Detection System github

- Developed a full-stack IoT dashboard using Python and FastAPI to capture and visualize live hardware sensor data streamed via MQTT from an ESP32 microcontroller.
- Engineered a custom authentication system backed by a relational MySQL database, implementing bcrypt for secure password hashing and UUIDs for persistent, cookie-based session management.
- Designed a containerized deployment pipeline using Docker and Docker Compose to reliably orchestrate the backend web server, database, and hardware communication channels.

Skills

Languages: C, C++, SystemVerilog, MATLAB, Python, Javascript, HTML, CSS, PyTorch, Docker, MySQL
Software: LTSpice, ModelSim, Intel Quartus Prime, Fusion360, Git, Linux/Unix, Cadence Virtuoso
Tools/Hardware: Oscilloscope, Multimeter, Function generator, Power supply, Soldering, ESP32, Arduino